

```
GCM_MODE {
    unique_ptr<CTR_BE> m_ctr;
    unique_ptr<GHASH> m_ghash;
}
```

```
SymmetricAlgorithm::set_key(key)
sym_algo.h
```

```
GCM_Mode::key_schedule(key)
gcm.cpp
```

Implementation

```
m_ctr->set_key(key)
m_ctr->set_iv(zeros.data(), zeros.size())
m_ctr->encipher(buf)
m_ghash->set_key(buf)
```

```
StreamCipher::cipher(in, out, len)
stream_cipher.h
```

```
StreamCipher::cipher_bytes(in, out, len) = 0;
stream_cipher.h
```

```
CTR_BE::cipher_bytes(in, out, len)
ctr.cpp
```

```
AEAD_Mode::set_associated_data(ad)
aead.h
```

```
AEAD_Mode::set_associated_data_n(idx, ad) = 0
aead.h
```

```
GCM_Mode::set_associated_data(idx, ad)
gcm.cpp
```

```
GHASH::set_associated_data(input)
ghash.cpp
```

```
Cipher_Mode::start(nonce)
cipher_mode.h
```

```
Cipher_Mode::start_msg(nonce, len) = 0
cipher_mode.h
```

```
GCM_Mode::start_msg(nonce, len)
gcm.cpp
```

Implementation

```
m_ctr->set_iv(y0.data(), y0.size())
m_ctr->encipher(y0)
m_ghash->start(y0)
clear_mem(y0)
```

```
Cipher_Mode::process(msg)
cipher_mode.h
```

```
Cipher_Mode::process_msg(msg, len) = 0
cipher_mode.h
```

```
GCM_Encryption::process_msg(buf, sz)
gcm.cpp
```

Implementation

```
m_ctr->cipher(in = buf, out = buf, size)
m_ghash->update({buf, sz})
```

```
Cipher_Mode::finish(block, offset = 0)
cipher_mode.h
```

```
Cipher_Mode::finish_msg(block, offset) = 0
cipher_mode.h
```

```
GCM_Encryption::finish_msg(buffer, offset)
gcm.cpp
```

Implementation

```
m_ctr->cipher(in = buf, out = buf, size)
m_ghash->update({buf, sz})
m_ghash->final(mac)
buffer.insert(mac) // insert mac tag at end
```